Christian Richardson

AI/ML Engineer

Kissimmee, FL (+1 407 863 2776)
christianrichardson.finance@gmail.com
Linkedin • Github • Stackoverlfow • dev.to

Summary

Senior AI/ML Engineer with **10+ years** of experience building production AI systems across healthcare, finance, and enterprise applications. Expertise in **LLMs, Agentic AI, RAG, Speech AI, Computer Vision, and Generative AI**, with hands-on experience in LangChain, Hugging Face, PyTorch, FastAPI, Docker, Kubernetes, and AWS. Proven track record of designing scalable AI platforms, fine-tuning foundation models, optimizing inference pipelines, and delivering cloud-native AI solutions that serve real-world, high-concurrency production workloads. Strong collaborator and technical mentor with a passion for transforming cutting-edge AI research into reliable, business-impacting products.

Skills

- AI & Generative AI: LLMs, Agentic AI, AI Agents, RAG, Conversational AI, Prompt Engineering, LangChain, LangGraph, MCP, LlamaIndex
- LLM Platforms: OpenAI GPT, Anthropic Claude, Google Gemini, Hugging Face, Mistral, AWS Bedrock, Vertex AI, Azure AI
- AI Coding Assistants: Claude Code, Cursor AI, GitHub Copilot, OpenAI Codex
- Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LoRA, QLoRA, DeepSpeed, Fine-Tuning, Distributed Training
- Speech AI & NLP: Whisper, XTTS v2, F5-TTS, ASR, TTS, Speaker Diarization, Speech Separation, NLP
- Computer Vision: YOLOv8, ResNet, OCR, Vision-Language Models (VLMs), OneFormer, FCOS3D, BEV, ONNX
- Backend: Python, FastAPI, Flask, Node.js, Java, REST APIs, SQLAlchemy, Microservices
- Data & Vector Databases: PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Pinecone, FAISS, ChromaDB, Weaviate
- Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Kafka, RabbitMQ, OpenTelemetry, LangFuse, GitHub Actions, Jenkins
- Data Engineering: Spark, PySpark, Hadoop, Databricks, Big Query, ETL/ELT
- Tools: Git, Jira, Notion, Zapier, Make, n8n, Agile, Technical Leadership, Mentorship

Experience

Ellipsis Health, California, United States
Senior AI Engineer 03/2024 – Present

Built **SAGE**, an AI-powered conversational agent platform enabling real-time mental health signal detection, patient care management, and intelligent clinician routing via voice-based interactions—serving as the primary patient engagement interface for proactive healthcare delivery.

- Built end-to-end voice AI pipelines integrating **Twilio, Whisper, OpenAI GPT, ElevenLabs, LangChain**, and multilingual TTS for natural multi-turn healthcare conversations.
- Developed speech AI pipelines featuring **speaker diarization, speech separation, multilingual TTS fine-tuning (XTTS v2, F5-TTS),** and data augmentation to improve transcription accuracy and speech naturalness.
- Designed agentic AI workflows using **LangChain, LangGraph, MCP, RAG, and Pinecone**, enabling autonomous tool calling, clinician lookup, scheduling, and contextual reasoning.
- Built **OCR** and Document Intelligence pipelines using vision-language models and LLMs to extract structured clinical data from medical documents.
- Developed scalable AI backend services with **Python, FastAPI, SQLAlchemy, PostgreSQL, Docker, and Kubernetes**, supporting high-concurrency production workloads.
- Implemented production LLM observability using **LangFuse**, monitoring latency, hallucinations, prompt versions, costs, and conversation quality while enforcing healthcare safety guardrails.
- Delivered healthcare-compliant safety and privacy controls, including role-based access, PII/PHI-aware conversation handling, guardrails for sensitive topics (crisis detection, self-harm language), safe fallback responses, and audit logging—ensuring **HIPAA** alignment and patient trust in AI-driven care interactions.
- Mentored junior developers and led knowledge-sharing sessions, strengthening team capability in LLM application development, agent orchestration, and production reliability practices.

Bittensor Toronto, Canada

Senior AI Engineer 10/2021– 02/2024

Actively participated in open-source development, sharing insights and code that contributed to the growth and success of the several Bittensor, the most well-known leading decentralized AI network, subnets working closely with several subnet owners.

- Contributed to decentralized AI subnets while developing scalable AI inference and distributed training infrastructure.
- Built custom LLM fine-tuning pipelines using **Hugging Face Transformers, LoRA, DeepSpeed**, and optimized inference with ZeRO-3, GPU acceleration, quantization, and distributed NVIDIA GPU training.
- Built large-scale ML data pipelines using **Spark, Hadoop, Databricks, NumPy, and Pandas** for healthcare, finance, and legal AI applications.
- Developed production APIs for 2D-to-3D reconstruction (Trellis), image recommendation systems, FAISS vector search, and computer vision inference.
- Built event-driven AI microservices using **Kafka, RabbitMQ, Azure SignalR, and REST APIs**, processing millions of daily AI events.
- Automated AWS infrastructure using Lambda-based scheduling to reduce cloud costs and improve resource utilization.
- Delivered data engineering and ML workflows on **Databricks** (Spark-based), enabling collaborative notebooks, scheduled jobs, and production-grade pipelines for feature generation, training, and evaluation.
- Orchestrated infrastructure-as-code deployments using **Docker**, **Terraform**, **AWS EC2, EKS** and **GCP**, enabling CI/CD, OTA updates, and scaling applications to 50K+ users**.**

Net Solutions, California, United States

Full Stack Developer | AI/ML Engineer 06/2016 – 09/2021

Developed AI-powered web platforms across healthcare, automotive, manufacturing, retail, and eCommerce using React,Python, and FastAPI.

- Designed and deployed computer vision solutions using **YOLOv8, ResNet, OCR, segmentation models,** and visual similarity search for industrial inspection and document intelligence.
- Built 3D computer vision and perception systems including **2D-to-3D reconstruction, FCOS3D, BEV perception, OneFormer segmentation,** and lane detection for ADAS applications.
- Optimized deep learning models through **quantization, pruning, ONNX conversion,** and knowledge distillation for edge deployment and C++ inference.
- Developed image-processing pipelines for segmentation, landmark detection, color normalization, and feature extraction.
- Collaborated with cross-functional engineering, product, and data science teams while mentoring junior engineers and promoting engineering best practices.
- Designed and deployed 10+ React/Next.js and Node.js platforms, integrating Python-based ML pipelines for analytics, personalization, and user behavior insights.

Education

Bachelor's Degree in Computer Science, Brooklyn College

Brooklyn, NY, United States (03/2012–05/2016)